3 Project Description and Construction Methodologies

3.1 Need of the Project

3.1.1 Territory-Wide Housing Issues

3.1.1.1 Housing has been one of the focuses in the Policy Addresses for last few years. According to the statistics in 2011, there are currently 2.6 million residential units in Hong Kong, accommodating 2.35 million households. Of those, 730,000 households live in Public Rental Housing (PRH) and 380,000 in self-owned units acquired with government subsidies. In other words, almost half of the households in Hong Kong are benefiting from some form of housing subsidy by the Government. Of the remaining 1.24 million households living in private properties, 870,000 are owner-occupants. Taken together, nearly 85% of households live in PRH units, subsidised home ownership scheme flats or their own private properties. Hence, housing has been one of the focus areas in the Policy Addresses for last few years.

3.1.1.2 The 2011-2012 Policy Address had identified that maintaining PRH production, and increasing land supply for subsidised sale flats and private housing would be the key to tackle issues relating to housing. According to the Public Housing Construction Programme of the Housing Authority (HA), about 75,000 PRH units would be completed in the next five years. In terms of land supply, the government set a target for an average of 20,000 private residential flats each year in the next decade in order to ensure the healthy and stable development of the property market. The Policy Address had mentioned various measures to expand the land resources and exploring the possibility of converting agricultural land in North District and Yuen Long currently used mainly for industrial purposes or temporary storage, or which was deserted into housing land was one of them.

3.1.1.3 Since the 2013 Policy Address, the Government has adopted a multi-pronged approach to increase land supply in the short, medium and long terms including the review of deserted agricultural land mentioned in paragraph 3.1.1.2 above.

3.1.1.4 The 2014 Policy Address stated that the Government accepted the recommendation of the Long Term Housing Strategy Steering Committee that public housing should account for 60% of the 10-year housing supply target of 470,000 units. Among public housing, the supply of PRH should be 200,000 units while that of subsidised sale flats 80,000 units.

3.1.1.5 In the 2015 Policy Address, the Chief Executive announced the development of brownfield sites in HSK NDA, YLS and New Territories North as a measure to increase land supply. The Government is studying feasible measures, including accommodating some of the users in multi-storey buildings and taking the "brownfield sites" inside the HSK NDA as a pilot case, with a view to taking forward the development of NDAs.

3.1.1.6 Both the 2016 and 2017 Policy Addresses have stated that the development of YLS is medium to long term land supply measures. In the 2017 Policy Address, the Government is determined to implement as soon as possible a number of large-scale projects including NDAs and the extension of new towns and the YLS development is one of them. These NDAs and new town extensions would provide additional

housing, community facilities, public space and land for economic use. These development projects are expected to provide close to 200,000 housing units and over 8.6 million square metres of industrial and commercial floor area between 2023 and 2038.

3.1.1.7 Through the YLS development, development potential of the degraded brownfield could be released to meet the territory's housing need. The YLS development, as one of the major sources of housing land supply, would provide about 28,500 new flats, of which 60% is public housing (about 17,300 units), accommodating a total population of about 88,000. The storage and workshop, commercial and community uses would generate about 10,500 employment opportunities. The development would provide the supporting commercial and community facilities to serve the YLS population and the residents of the adjacent existing villages. Commercial floor space has been reserved in the residential sites and the "Other Specified Uses)" annotated "Mixed Use" developments for retail and restaurant uses. Various community facilities including clinic, sports centre, community hall, schools, kindergarten, police station and fire station cum ambulance depot are planned to serve the communities in the area.

3.1.2 Optimising the Development Potential of Brownfield Sites

3.1.2.1 In Hong Kong, brownfield sites generally refer to the agricultural land in the rural New Territories that has been converted for general open storage and rural workshop uses, especially the relatively flat and more accessible areas. These sites have been occupied by various haphazard industrial operations such as open storage, port-backup facilities and rural industrial workshops, which are often incompatible with the surroundings and cause industrial/residential interface issues. Development of brownfield sites provides an opportunity for optimisation of the utilisation of the scarce available land resources.

3.1.2.2 Of the 223.5 ha of land, about 100 ha in YLS is currently occupied by brownfield operations of various types including open storage, warehouses, waste recycling workshops, vehicle repair workshops, etc. Through consolidation of the brownfield operations in into multi-storey buildings or through other efficient means of land use that is being explored taking HSK NDA as a pilot case, development of brownfield sites could be released for more other uses such as housing to address the territory's housing needs.

3.1.3 Improvement of Rural Environment

3.1.3.1 YLS PDA is currently rural in character with a mixture of land uses. The predominant uses are brownfield operations including open storage yards, warehouses, workshops, industrial operations. These brownfield operations are intermingled with rural settlements and residential developments, agricultural land, livestock farms and vacant land. Proliferation of brownfield sites has resulted in degradation of the rural environment due to the increasing noise nuisances to surrounding residents, and discharge of industrial waste water, i.e. from daily cleaning at the working area, waste water from workshops after heavy rainfall etc.. Furthermore, any of these industrial premises which may not be properly connected to sewer system could result in discharge of untreated sewage into nearby water bodies and hence deteriorate water quality.

3.1.3.2 Through the YLS development, the brownfield operations are consolidated in the multi-storey buildings of the Employment Belt located adjacent to the YLH. Road

network is specially planned to facilities traffic from the Employment Belt to enter YLH directly. This could avoid freight traffic travelling through the residential communities in PDA as the existing villages, alleviating the industrial/residential interface. Besides, the proposed environmental initiatives including revitalisation of nullahs, preservation of watercourses of higher ecological value, secondary woodland and active agricultural land and the proposed greening and landscaping in open space, amenity areas and along the hillside river corridor would enhance the overall environment.

3.1.4 Major Connection in the District

3.1.4.1 Yuen Long New Town has been developed into a major regional centre with provisions of commercial and community facilities in the Northwest New Territories (NWNT), whereas Tin Shui Wai has been developed another new town in the NWNT. With the completion of the public transit connections such as YLH, Route 3, the Light Rail Transit (LRT) network and Mass Transit Rail (MTR) West Rail network, the linkage of the PDA with the main urban area has been improved. The Study has proposed a comprehensive road, pedestrian and cycling network linking the PDA with the strategic roads, LRT and MTR West Rail stations.

3.2 Appreciation of Existing Environment

3.2.1.1 The PDA is located at south side of YLH and Yuen Long New Town. As discussed above, the PDA comprises of 3 areas (i.e. Area 1, Area 2 and Area 3). Other than the PDA, the Project also includes a number of associated infrastructure consisting of works for roads, water supplies and sewer pipes and a service reservoir outside the PDA. **Figure 1.1** shows the locations of the 3 Areas and associated infrastructure. The following sections provide an overview of the existing environment of these areas and infrastructure.

Area 1

3.2.1.2 Area 1 is located in TYST and connected to the YLH. Existing land uses are mixed with brownfield sites for various uses such as open storage and rural industrial workshop, etc. Two concrete batching plants are located in the northern part of Area 1 and some container storage sites are located in the south-western part, along Long Hon Road, of the area. The structures within these brownfield sites are mainly temporary structures of various forms.

3.2.1.3 Within Area 1, there are residential developments along Sha Tseng Road and TYST Road such as Recours La Serre, The Eldorado, Kisland Villa, etc. A non-indigenous village, Sha Tseng Tsuen, is also located in Area 1. These residential developments are located away from YLH and hence not badly affected by the highway's traffic noise and vehicular emission. All these residential developments in Area 1 are low density of about 1 to 4 storeys but some of them are located very close to existing brownfield sites and hence are affected by those brownfield operations to a certain extent. Other than brownfield sites and low density residential developments, there are also woodland and agricultural land with watercourses of different sizes and qualities running in between these existing developments. Some woodland is close to the Tai Lam Country Park to the south west and forms a continuous resources in terms of visual and ecology. There is also one graded historic building, i.e. Yeung Hau Temple in TYST, within Area 1, which is culturally related to the historical development of the indigenous villages.



Yeung Hau Temple



Active Agricultural Land

Area 2

3.2.1.4 Area 2 is located to the east of Area 1 along Kung Um Road and in the vicinity of Lam Tei East Road and Lam Tei West Road. There are a number of brownfield sites with rural industrial uses such as warehouses, open storage, vehicle maintenance, waste recycling, etc. A non-indigenous village, i.e. Tin Lung Tsuen, is located in the northern part of this area. The village is far away from YLH and thus not badly affected by the highway's traffic noise and vehicular emission. The separation distance between some brownfield sites and residential sites could be very close.

3.2.1.5 A number of village houses are built alongside Kung Um Road and Kiu Hing Road overlooking onto Yuen Long Nullah. Some of these houses are within a few metres from the existing road kerb line and adversely affected by the road traffic noise and vehicular emission.

3.2.1.6 There is an active egretry near Pak Sha Tsuen with small patches of secondary woodland located to the south of Area 2 outside the PDA boundary. There are also a few watercourses of different sizes and qualities within the PDA. Yuen Long Nullah is running from south to north between Kung Um Road and Kiu Hing Road in this area. No graded historic building is present within Area 2.

3.2.1.7 There are one chicken farm and one pig farm within Area 2. The nearest residential developments, Sun Mei Garden and Pak Sha Tsuen, are outside the PDA but located only about 50-100m away from these livestock farms. It is anticipated that these residential developments would be affected by the odour from the operation of these livestock farms, especially during hot summer time with unfavourable weather conditions.



Yuen Long Nullah



Tin Lung Tsuen

Area 3

3.2.1.8 Area 3 is located south of Area 2 along Kung Um Road, at the most southern part of the PDA. The area is densely built of temporary structures for brownfield operation uses. Residential developments include Pak Sha Tsuen, Golden Villa and Wong Nai Tun Tsuen, etc. are located near the PDA and are mainly typical village houses of up to 3 storeys high. Yuen Long Nullah runs generally from south to north conveying stormwater runoff from Tai Lam Country Park to the downstream receiving water bodies. Yuen Long Nullah is an important visual element for the community in the vicinity.

3.2.1.9 There are some individual houses scattered in Area 3. A large piece of secondary woodland is present to the western part of Area 3 and a few water courses of different sizes and qualities are found within Area 3. Tai Lam Country Park is located to the south of the area. No graded historic building is present within Area 3.

3.2.1.10 There are currently two chicken farms and two pig farms within Area 3. The nearest residential developments are at Wong Nai Tun Tsuen, Tai Tong Tsuen and Pak Sha Tsuen at approximately 400m away from these livestock farms. Given these four livestock farms are in the vicinity of the villages, it is anticipated that these residential developments would be affected by the odour from those livestock farms' operation, especially during hot summer time with unfavourable weather conditions.



Proliferation of Brownfield Sites



Livestock Farm

Associated Infrastructure

3.2.1.11 There are supporting infrastructure / road improvement works outside the PDA boundaries to support the proposed housing development. New roads include slip roads for connection to Pok Oi Interchange and a new PTI next to Yoho Midtown. The slip roads will divert PDA traffic away from the busiest section of the YLH as well as allow connectivity to West Rail Yuen Long Station. Improvement to existing road outside the PDA include modification works to TYST Interchange and the TSWW Interchange for enhanced connection to YLH, as well as some local road widening works to improve road safety and pedestrian environment.

3.2.1.12 Besides road works, a new sewer would be required to convey some of the TSE from the proposed YLS STW to the existing inlet chamber of the NWNT tunnel at San Wai discharging to the Urmston Road submarine outfall. This new sewer would run along the planned roads within the YLS PDA and existing roads including Hung Tin Road, Ping Ha Road, Tin Ha Road and Ha Tsuen Road outside the PDA.

3.3 Consideration of Scenario Without the Proposed Development

3.3.1.1 Without the Project, alternative housing sites may need to be identified on alternative greenfield sites, which generally refers to previously undeveloped land. As mentioned in **Section 3.1.2**, in Hong Kong, land supply has been a subject of concern for many years. In order to accommodate the increasing population and household growth effectively, it would be more effective to develop on previously developed land instead of looking for scarce greenfield land.

3.3.1.2 From urban planning perspective, it is more effective to develop this area as it was close to the well-established areas as well as new planned development areas such as Yuen Long New Town, Tin Shui Wai New Town and the future HSK NDA. For Yuen Long New Town in particular, it is a civic and commercial hub in Yuen Long District with well-developed public transport infrastructures such as LRT stations and MTR West Rail Stations. Tin Shui Wai New Town located to the northwest of the YLS PDA, comprises a Wetland Park for various recreational and educational purposes. The future HSK NDA is proposed to be developed to meet long-term housing, economic, social and environmental needs. Summarizing the above, the YLS PDA is located at an area which can create a synergy effect with these new towns by better utilisation of existing facilities, and at the same time providing future facilities to fulfil inadequacies and support future development needs of the wider community.

3.3.1.3 Other than supplying land for addressing territory-wide land supply, Stages 1, 2 and 3 of CE exercise conducted for this study have collated views and opinions from various stakeholders. The CE exercise have also revealed a number of aspirations from the local communities that cannot be fulfilled without implementing the Project, including:

- Road improvement works within the PDA such as Kung Um Road and Kiu Hing Road which are mostly sub-standard;
- Enhancement on the walking and cycling network and the increase in provision of public transport at YLS;
- Beautification and revitalisation measures for the Yuen Long Nullah;
- Upgrading works on the drainage and sewage treatment facilities; and
- Protection of natural resources within the PDA.

3.4 Recommended Outline Development Plan

3.4.1.1 The RODP adopted for this EIA Study is shown in **Figure 3.1**. The following sections present the design principles and the key planning elements proposed in the RODP.

3.4.1.2 The planning of YLS has taken account of the existing development characters and integration with Yuen Long New Town in its north and the rural and natural landscape in the surroundings. The YLS development would provide about 28,500 new flats for a total population of about 88,000. In line with the Long Term Housing Strategy, the housing mix of 60% for public housing and 40% for private housing is adopted. Economic uses including commercial, storage and workshop uses have been planned for employment opportunities. Together with the supporting infrastructure, commercial, social and community facilities planned, the YLS

development would be developed as the new town extension for creation of a balanced community.

3.4.2 Key Planning and Urban Design Principles

3.4.2.1 The overall planning and urban design framework, which were adopted for the RODP, are briefly described as below.

Accommodating Diversity for Housing, Storage and Workshop, and Preservation of Natural and Landscape Resources

3.4.2.2 Within the PDA, 5 Planning Areas are proposed. Three residential communities are planned with different development intensities taking account of the existing and surrounding development characters. Each community is anchored with a key activity node comprising of a PTI, and commercial and community uses. The Employment Belt is planned for storage, industrial workshop and open storage uses, and is located at the northern part of TYST close to the highway for easy access to YLH. The freight traffic would not need to travel through the residential neighbourhood in the PDA. A Green Zone is designated at the southern part of TYST to preserve the existing largest contiguous piece of active farmland. Besides, watercourses of higher ecological value and secondary woodland would also be preserved as far as practicable.

Respecting Existing Urban and Rural Environment by Stepped Building Heights

3.4.2.3 Through the tapering of development intensities and building heights from north to south, YLS will integrate with its surrounding settings by high density developments in the north near Yuen Long New Town and transitioning to medium and low-density developments in the south to blend in with the rural setting and Tai Lam Country Park. Low-density developments are planned in the TYST area to complement and integrate with existing low-density and low-rise residential developments.

3.4.2.4 The urban to rural qualities are further enhanced with the provision of Scenic Cycling Tracks along the Hillside River Corridor and retaining active agricultural land within the PDA. Bicycle parking facilities are planned near the proposed PTIs, and a network of roadside cycle tracks is widely available within the urban streetscape connecting the residential neighbourhoods.

Blue-green Network

3.4.2.5 The PDA is framed with a setting of natural and rural environment together with the presence of various natural landscape and watercourses. These resources form parts of the urban structures for creating a liveable, sustainable and green environment. These blue and green resources are combined to form the blue-green network (the "blue" composed of watercourses of higher ecological value, revitalised nullahs and the new Hillside River Corridor and the "green" comprised of the open space, the active agricultural land in the Green Zone, the secondary woodland and the amenity areas). These features are interwoven and integrated with the pedestrian and cycling track networks to enhance the enjoyment of walking / cycling by providing rural and natural landscape alongside. The blue-green network shapes and gives character and identity of YLS.

3.4.2.6 The "blue" features provide opportunities for the formulation of sustainable water management measures including the STW with tertiary treatment level, reedbed/ retention pond and flood retention facilities.

Creating Open Space and Green Network

3.4.2.7 Open spaces of various sizes and functions are planned and the residential communities will be served with district open space. Together with the preserved active agricultural land, secondary woodland, watercourses, as well as revitalised nullahs and new water channels, all these will form a comprehensive green landscape network to meet leisure and recreational needs while creating identity and character for YLS. These features are integrated with each other in the overall landscape framework.

3.4.2.8 Open space is conveniently accessible not only for the future population but also existing residents and villagers in and near the PDA. Open space corridors (zoned "Local Open Space" ("LO")) also function as buffer between the existing low-rise villages and planned high-rise public and private developments to address concerns of local villagers on the potential wall effect due to new high rise developments. In addition, low-building areas have also designated in the new residential areas next to existing villages.

3.4.2.9 A major Green Zone is designated in the centre of the PDA by combining secondary woodland and the largest piece of contiguous active agricultural land. This zone reflects the rural characteristic of the area.

3.4.2.10 The backdrop of Tai Lam Country Park also forms part of the green network through the tapering of building heights, creation of visual corridors and linkages.

View Corridors and Breezeways

3.4.2.11 When formulating land use proposals, the view corridor at Yuen Long Park Aviary Pagoda to the north of PDA has been considered. The gradation of development intensity and building heights allow developments to optimise views to the mountainous backdrop of the Tai Lam Country Park. Another key view corridor is along the revitalised Yuen Long Nullah.

3.4.2.12 New roads within the PDA are proposed to mainly align with the prevailing wind direction as far as possible. Furthermore, stepping building height approach has been adopted to help wind deflection and avoid air stagnation. The gradation of building height profiles has been studied in relation to areas with different plot ratios tapering downwards in the southbound direction.

Improving Connectivity

3.4.2.13 The connectivity of YLS with Yuen Long New Town and HSK NDA will be fully enhanced through the planning of a new and comprehensive road, pedestrian and cycling networks. Reserve has been provided in the RODP for the potential extension of the environmentally friendly transport services (EFTS) from the HSK NDA for providing efficient and comfortable transport between the two developments.

Revitalisation of Nullahs

3.4.2.14 The existing nullahs are key assets of the PDA and will be revitalised as far as practicable to reach their full potential as an attractive asset of the PDA. The nullahs will provide a rural and natural sensation in the urban context and act as key north-south visual corridors.

3.4.2.15 In line with recent advocacy for integrated green infrastructure systems and aspiration to improve aesthetics and ecological value of the land, a green and eco-

hydraulics approach will be applied. A key consideration is to improve aesthetic whilst maintain and compensate for the hydraulic performance of the channel. Some of the green and eco-hydraulics approaches which will be considered are:

- Using grasscreting and extensive vegetation embankment, e.g. nullahs in Jordan Valley and Tsui Ping;
- Using gabions and geo-fabric reinforced grass lining, e.g. Yuen Long Bypass Floodway;
- Using unlined channel beds such as rip-rap base and retaining meanders, e.g. Kam Tin River, Shan Pui River and Ng Tung River; and
- Creating shallow ponds, wetland habitats and reedbeds, e.g. Yuen Long Bypass Floodway and San Tin.

Preservation of Egret Flight Path

3.4.2.16 The active egretry near Pak Sha Tsuen is an important ecological asset to the PDA. The design of RODP has considered the preservation of egrets' flight path.

3.4.2.17 A 90m to 135m wide district open space is designated between sites R2.6a, R2.6b and R2.6c to protect the flight corridors of the egrets, which may be impeded by the residential buildings, by retaining the green linkage to the exiting foraging round at retained farmland at Shan Ha Tsuen.

3.4.2.18 An about 50m wide non-building area (NBA) is designated at Sites R2.6d to protect the fight path in East-to-West direction with the proposed new foraging ground at Retention Lake.

3.4.2.19 A 30m wide NBA is designated between the two "R2(SSF)c" sites and a Low building area (LBA) is also designated at the northeast corner of site OU3.2a to protect the flight path in north-to-south direction and to provide a flight corridor leading to the fully revitalised Yuen Long Nullah at the southern end of PDA.

3.4.3 Key Development Parameters and Land Use Budget

3.4.3.1 The current development proposals constitute residential use, storage and workshop uses, government, institution or community (GIC) use, as well as agriculture, open space, amenity and green belt. The land use budget of the RODP is summarized in the following table and the development parameters for the PDA are set out below:

Table 3.1 Key development parameters under RODP

Total Population	**About 88,000 (including the population of about 2,400 of the existing/planned residential estates and non-indigenous villages that are proposed to be retained)**
Population (new)	85,400 (Public: 53,100; Private: 32,300)
No. of Flat (new)	28,500 (Public: 17,300; Private: 11,200)
Housing Mix • Housing Units • Population • Housing Land	 Public 61% : Private 39% Public 62% : Private 38% Public 32% : Private 68%
Plot Ratio for Residential • Public • Private	 4 and 5 1, 1.5, 2.4, 4 and 5

Table 3.2 Land Use Budget

Land Use	Proposed Zoning on RODP	Area (ha) (% of total)
(A) Residential		***72.5(32.4%)***
- Subsidised Housing	"RSc", "R1(SSF)c", "R2(SSF)c"	18.9
- Private Housing	"R1c", "R2c", "R2", "R3", "R4", "R5", "R5(EDA)"	46.3
- Village Resite	"R6"	1.3
- Other Specified Uses (Mixed Use)	"OU(MU)"	5.9
(B) ***Storage and Workshop***		***11.1 (5%)***
- Other Specified Uses (Open Storage)	"OU(OS)"	1.4
- Other Specified Uses (Storage Use)	"OU(SU)1", "OU(SU)2"	8
- Other Specified Uses (Storage and Workshop)	"OU(S+W)"	1.7
(C) ***Government, Institution or Community***		***10.2 (4.5%)***
- Government	"G"	9.2
- Institution or Community	"IC"	1
(D) Education	"E"	***5.7 (2.6%)***
(E) Public Utilities[1]		***15.3(6.8%)***
Other Specified Uses (Sewage Treatment Works)	"OU(STW)"	4.0
Other Specified Uses	"OU"	1.1
Other Specified Uses (Hillside River Corridor with Scenic Cycle Track) [4]	"OU(Hillside River Corridor with Scenic Cycle Track)"	10.4
(F) Road and River Channels		***53.5(22.2%)***
(G) Amenity	"A"	***6.5(2.9%)***
(H) Open Space		***28.6(12.8%)***
- District Open Space	"DO", "DO(1)"	15.7
- Local Open Space	"LO", "LO(1)"	12.9
(I) Agriculture[2]	"AGR"	***10.5(4.7%)***
(J) Green Belt[3]	"GB", "GB(1)"	***9.6 (4.3%)***
(K) Village Type Development	"V"	***0.1(0.04%)***
Grand Total		**223.5**

Notes:
(1) "OU(Service Reservoir)" (about 0.6ha) is outside the PDA boundary
(2) "AGR" zones include the preserved active agricultural land and one existing chicken farm located at the southern end of Kung Um Road.
(3) "GB" zones cover the preserved ecologically important areas including secondary woodland, watercourses of higher ecological value and buffer areas.
(4) "OU (Hillside River Corridor) with Scenic Cycle Track" also includes reedbed/retention pond.

3.4.4 Residential

3.4.4.1 Under the RODP, the approximate 28,500 new residential will house an estimated new population of approximately 85,400 persons. With the existing population within the Project Area, the overall population is projected to be around 8,8000 persons upon full development. The public to private housing mix is about 61:39.

3.4.5 Storage and Workshop

3.4.5.1 Taking advantage of the close connection with the highway network, about 11.1 ha of land at the south of YLH in the PDA at TYST area for storage and workshop uses. Through modifying the TSWW and TYST Interchanges particularly the former, more traffic movements are allowed to cater for direct access of freight traffic to

YLH. These sites will provide about 389,000 GFA for storage and workshop uses and reserved about 1.4ha for open storage uses.

3.4.6 Government

3.4.6.1 "Government"f ("G") zones are intended for the provision of G/IC facilities such as police station, fire station, hospital, sports centre, social welfare building, etc, serving the needs of the local population of about 88,000 and complementing provisions in the wider area of TYST and Tai Tong. It is also intended to provide land for uses directly related to or in support of the work of the Government or organisations that provide social services to meet community needs.

3.4.7 Institution or Community

3.4.7.1 Three sites (Site G1.1, 1.3 and 1.4) are designated as "Institution or Community" ("IC") on the RODP to preserve the existing religious institutions

3.4.8 Education

3.4.8.1 Schools will be provided within the "Education" ("E") zone within the Project area. In general, developments at these sites are subject to a maximum building height of 8 storeys.

3.4.9 Public Utilities

Other Specified Uses (Sewage Treatment Works) ("OU(STW)")

3.4.9.1 An "OU(STW)" zone (Site OU3.3, at the southern end of Kung Um Road is reserved for the construction of an aboveground STW for treatment and disposal of sewage arising from the PDA. A reedbed/ retention pond with an area of around 4ha is proposed adjacent to the "OU(STW)" zone to provide biological treatment and TSE polishing, subject to further detailed study. It is intended to minimise the pollution loading being discharged from the YLS development. Due to the uncertain pollution removal efficiency of the reedbed, its polishing function would only be regards as an extra provision and would not be considered in the pollution load offset balance as stated in **Chapter 7**. Reuse of TSE as reclaimed water for non portable use including toilet flushing would be implemented in the PDA and existing and future developments.

Other Specified Uses (Refuse Collection Point and Sewage Pumping Station) ("OU(RCP&SPS)")

3.4.9.2 Two "OU(RCP&SPS)" zones (Sites OU1.12 and OU2.2) are intended for the sewage pumping stations (SPS) and refuse collection points (RCP).

Other Specified Uses (Electricity Substation) ("OU(ESS)")

3.4.9.3 Site OU2.1 (site area of about 0.3 ha) is intended for the construction of an Electricity Substation.

Other Specified Uses (Service Reservoir) ("OU(Service Reservoir)")

3.4.9.4 Site OU1.12 (site area of about 0.6 ha) located to the south of the PDA at TYST area is intended for a potential reclaimed water service reservoir.

Other Specified Uses (Hillside River Corridor with Scenic Cycle Track) ("OU(Hillside River Corridor with Scenic Cycle Track)") – 9.8ha

3.4.9.5 About 10 ha of land along the western boundary of *LOHAS Living* is zoned "OU(Hillside River Corridor with Scenic Cycle Track)", including land for reedbed/ retention pond. In order to mitigate the impact of additional surface runoff generated from the proposed development, the new drainage channel is proposed for water storage as well as its normal drainage function, which also provides opportunity for ecological enhancement. A scenic cycle track, footpath and landscaping are proposed alongside the drainage channel for public enjoyment.

3.4.10 Amenity

3.4.10.1 This zone is reserved along developments, roads and YLH to allow provision of roadside planting, buffer and footpath to enhance the environment and to serve as visual buffers.

3.4.11 District Open Space

3.4.11.1 A number of "DO" are planed within the Project area to provide recreational and leisure spaces. "DO" is intended primarily for the provision of outdoor open-air public space for active and/or passive recreational uses serving the needs of local residents as well as the general public. A network of district and local open space will create a naturalistic environment and green setting of the whole area. The DO sites will also serve as the flight corridor to protect the egrets' flight path.

3.4.12 Local Open Space

3.4.12.1 Local Open Space" ("LO") is provided throughout the Project area and is in close proximity to residential areas and employment centres to serve local residents and workers.

3.4.13 Agriculture

3.4.13.1 This zone is intended primarily to retain and safeguard existing quality agricultural land for agricultural purposes. The southeastern portion of TYST is predominately dry active agricultural land intermixed with wet and inactive agricultural land. This area is designated as "AGR" to allow the continuation of farming practices. The retained chicken farm at the southern end of the PDA is also zoned "AGR" (site AGR 3.1).

3.4.14 Green Belt

3.4.14.1 The planning intention of these zones is primarily for defining the limits of urban and sub-urban development areas by natural features and to contain urban sprawl. There is a general presumption against development within these zones.

3.4.14.2 Three land corridors adjoining the streams with ecological importance within the PDA are zoned "GB(1)" which intend to protect the habitats for wildlife from unauthorized encroachment and illegal dumping, while providing flexibility to the Town Planning Board to scrutinise developments under the planning permission system.

3.4.15 Village Type Development

3.4.15.1 The planning intention of "V" zone on RODP is mainly to retain existing houses at the boundary of PDA.

3.4.16 Road and River Channels

3.4.16.1 The road network consists of vehicular lanes, pedestrian footpaths, road-side amenity, cycle tracks, and road reserve for an EFTS, which would subject to further studies. A simple hierarchy of vehicular lanes is proposed to serve the PDA efficiently without resulting in excessively wide roadways.

3.4.16.2 A continuous cycle track designated as "Road" has been proposed along proposed residential developments in Area 2 to link with the cycle track network along the road.

3.4.16.3 There are three existing drainage channels within the PDA, namely Yuen Long Nullah, Tin Tsuen Channel and Yuen Long West Nullah, and decking of these channels would be avoided as far as practicable. Through revitalisation, these channels would provide a landscaped and pleasure walking environment for the community. In particular, the northern section of Yuen Long Nullah within Area 2 is proposed to be partially decked to provide adequate road space with potential for nullah revitalisation.

3.5 Proposed Construction Methodologies for Land Uses and Design

3.5.1 Site Formation

3.5.1.1 The site formation is to be carried out by phases. The order of magnitude of the inert Construction & Demolition (C&D) soft material generated from slope cut and formation is similar to the general fill formation required for design flood level. The fill material would be reused within the Project as far as practicable. Conventional earthwork cut and fill machineries will be used for the site formation.

3.5.1.2 Near the hillside, larger-scale slope cut and retaining wall is required to maximise the available land for development. At these locations, construction like soil nail installation and reinforced concrete retaining wall with temporary slope cut will be involved. At constrained locations, cantilever retaining wall like bored pipe wall is needed. This will be achieved by piling machinery.

3.5.2 Road Works and Utilities

3.5.2.1 The proposed road works and utilities include road networks, drainage systems, sewerage networks, water supply networks and utility construction to support the development in the new development area.

3.5.2.2 Except for the grade-separated interchanges, typical construction method for at-grade road construction would be adopted. The works involve earthworks, utilities laying, laying of sub-base materials and laying of bituminous or concrete surfacing layers.

3.5.3 Bridge Structures at Interchange

3.5.3.1 The substructures of the bridge structures at the interchanges will take the form of reinforced concrete (RC) columns & pile caps founded on RC bored piles.

3.5.3.2 Regarding superstructure, given the small total length, sharp road curves and complicated geometry of the bridge structures, it is envisaged that precast segmental construction and in-situ balanced cantilever will not be cost-effective and appropriate for this particular condition. Conventional in-situ bridge construction on formwork and falsework, with appropriate temporary traffic management, is envisaged to be adopted. The superstructure will take form of typical RC structures.

3.5.4 Underpass at Interchange

3.5.4.1 Cut-and-Cover method would be adopted for the underpass, which involving trench-excavation (i.e. open-cut) followed by in-situ construction of the tunneCHANNl structures.

3.5.5 Box Culvert / Bridge Crossing at Channel

3.5.5.1 Cut-and-Cover method would be adopted for the box culvert. The construction will have to be carried in particular seasons with less disturbance to the drainage flow. Temporary drainage arrangement will also be developed during the construction to avoid impact to the drainage.

3.5.6 Landscaping Works at Open Spaces, Amenity Areas and Slopes

3.5.6.1 Landscaping works at open space, amenity areas and slopes will be provided after site formation works and slope works. As these will mainly involve planting and minor pedestrian facilities, environmental impact is not anticipated.

3.5.7 Reclaimed Water Service Reservoir

3.5.7.1 The major construction works will include typical earthwork, concrete works for service reservoir structures, pipeworks and modification work to existing access road. Both cut-and-fill slopes shall be formed for the formation of platform for the service reservoirs construction and for the modification work to existing access road.

3.5.8 Noise Barriers/ Enclosures

3.5.8.1 Depending on the ground condition, the noise barrier/ enclosure height and the site constraint, the noise barrier/ enclosure will either be supported on shallow foundation or on piles. The noise panel is usually pre-fabricated and erected on site with appropriate temporary traffic management.

3.5.9 Building

3.5.9.1 The building in the new town will be constructed by the Housing Department, the Architectural Services Department or the private developers, depending on the building type and land use. It is envisaged that conventional building construction will involve piling work. For the superstructure, either precast block erection or in-situ concreting work is envisaged.

3.5.10 Sewage Treatment Works/ Sewage Pumping Stations

Sewage Treatment Works

3.5.10.1 A new YLS STW will be provided to handle the sewage arising from the new population and employment in the PDAs. The new YLS STW will be implemented in two phases to serve the different phases of development.

3.5.10.2 The new YLS STW (Phase 1) will start operation in Year 2030 to handle 1,800 m^3/day of sewage generated in the YLS Stage 1. The new YLS STW (Phase 2) will be expended to cope with the flow in subsequent stages, subject to review of the future population growth. The new YLS STW will be constructed with tertiary treatment for reuse of reclaimed water and secondary plus treatment (with UV disinfection and 75% nitrogen removal) for disposal of effluent with an ultimate design capacity of approximately 24,000 m^3/day.

3.5.10.3 The proposed YLS STW is a DP under Schedule 2. DP requiring EPs. Hence separate EIA Study for the proposed STW will be conduct by the future project proponent. Further details can be extracted in the Chapter 7 – Sewage and Sewerage Treatment.

Sewage Pumping Stations

3.5.10.4 The sewerage arrangement for the intermediate and ultimate development stages, include two intermediated SPSs which will be required to convey the sewage to new YLS STW. The two new SPSs will be implemented with pump capacity of 304 l/s located at the eastern side of TYST to pump the sewage collected in TYST to the eastern side of the PDA along Kung Um Road and pump capacity of 1,508 l/s located adjacent to Lung Tin SPS to pump the sewage to the YLS STW for treatment.

3.5.10.5 Construction of two intermediated SPSs will be carried out in the stage 2 in order to cope with the completion of stage 2 works for the Project.

3.5.10.6 Since these two SPSs are with an installed capacity of more than 2,000 m^3/day and are located within 150 m from existing and/or planned residential area or educational institution, these SPSs are classified as DPs under Item F.3, Part 1, Schedule 2 of the EIA Ordinance.

3.5.11 Revitalisation of Nullahs

Partial Decking and Revitalisation of Yuen Long Nullah

3.5.11.1 Partial decking and revitalisation of the Yuen Long Nullah will be provided along Kung Um Road and Kiu Hing Road to meet the traffic needs and as part of the urban design initiative for the YLS PDA, but it will reduce the capacity if the Yuen Long Nullah by reducing flow area and increase the roughness surface. A new drainage system that runs parallel with the Yuen Long Nullah is proposed (e.g. box culverts underneath Kung Um Road and Kiu Hing Road) to compensate the reduction flow areas and nullah capacity such that the concerned section of Yuen Long Nullah can still provide 50 year conveyance capacity. This portion of Yuen Long Nullah within YLS PDA will be provided greening and eco-hydraulics approach for rehabilitating the 12m to 16m wide trapezoidal concrete-lined Yuen Long Nullah (some 2.5km long).

Revitalisation of Existing Yuen Long (West) Nullah and Tin Tsuen Channel

3.5.11.2 As part of the urban design initiative, sections of the existing Yuen Long (West) Nullah and Tin Tsuen Channel, are proposed to be revitalised by various soft and hard landscape treatment proposals. Human access down to the revitalised channel is not proposed due to the inherited flood risk.

3.5.11.3 In line with the recent advocate of integrated green infrastructure system and aspiration to improve aesthetics and ecological value of the land, it is proposed to convert and rehabilitate the 9m wide rectangular concrete-lined Tin Tsuen Channel (some 1.5km long) and 11m wide trapezoidal concrete-lined Yuen long (West) Nullah (some 0.5km long). Similar to the Yuen Long Nullah, the Tin Tsuen Channel and Yuen Long (West) Nullah green and eco-hydraulics approach will be applied.

3.5.11.4 In addition, reference shall be made to DSD "Practice Note No. 1/2015 – Guidelines on Environmental and Ecological Consideration for River Channel Design" to tally with the concept of revitalising water-friendly culture in large-scale drainage improvement works and planning drainage networks for YLS PDA, so as to build a better environment for the public and balance between flood protection and ecological conservation.

3.5.12 Retention tank, Retention lake, Hillside River Corridor and Reedbed/ Retention Pond

3.5.12.1 The construction works include excavation of underground retention tank could be constructed by cut and cover method with associated shoring system, and the retention lake/ hillside river corridor as well as the reedbed/ retention pond could be constructed by open-cut method. The adopted construction methods are depended on the site constrains during the construction stage, which will be subject on various factors, such as topography and geology, etc. During the construction stage, the Contractor should require to provide any measures, such as sand traps, silt traps, sediment tanks, septic tanks, and soakaway system, etc, and get discharge licence from relevant authorities to ensure the surface runoff and sewage from construction activities do not cause adverse impact to the water bodies.

3.6 Consideration of Alternative Construction Methodologies

3.6.1 Alternative for Site Formation Levels

3.6.1.1 A majority of the PDA is on the Yuen Long Plain at approximately +5mPD, with elevation increasing in proximity to the mountain range with a maximum elevation of +56mPD. The site formation for the PDA will keep as close to the existing ground profile as possible. The accesses of the exiting village and the new development area will be considered to match the new infrastructure level.

Option 1: PODP Site Formation

3.6.1.2 In the preliminary site formation design based on the PODP, quantity of excavated material is greater than the required quantity of fill material which requires net export of a large amount of C&D material from the YLS development project.

Option 2: RODP Site Formation

3.6.1.3 Later in the amended site formation design based on the RODP, design refinement has been made in the cut and fill volume balancing. Particularly in phase 2, the site formation level is slightly raised to provide higher defence of flooding protection against climate change, while maximising the on-site reuse of fill material.

3.6.1.4 The comparison of cut and fill material generation for the PODP and RODP site formation designs are tabulated in **Table 3.4** below and the site formation layout plan for PODP and RODP are illustrated in **Appendix 3.1**

3.6.1.5 It can be seen that the Option 2 adopted has less surplus C&D waste material and hence is more environmentally friendly in term of waste disposal. It also slightly reduces the construction traffic to dispose of the surplus material.

Table 3.4 Comparison of Cut and Fill Material Generation for PODP and RODP Site Formations

Material	In-situ Volume (m^3)	
	Option 1: PODP Site Formation	**Option 2: RODP Site Formation**
Cut	1,690,000	1,620,000
Fill	1,580,000	1,540,000
Cut minus Fill	**110,000**	**80,000**

3.6.2 Alternatives for TSWW Interchange

Improvement Option 1

3.6.2.1 The layout of TSWW Interchange Improvement Option 1 is presented on **Appendix 3.2**. In this option, the main objective of the improvement work is to provide a local road connection between the Tin Shui Wai / Hung Shui Kiu areas and the YLS PDA via Hung Tin Road. There is no direct slip road connection between the YLS PDA and the YLH in this option. However vehicles from / to YLS PDA can indirectly U-turn at the diamond interchange on Hung Tin Road at the north and access to the YLH.

3.6.2.2 The off-slip road from YLH westbound to Tin Shui Wai has to be realigned to a depressed road / underpass, so as to provide grade separation with the local connection road between Tin Shui Wai and YLS. A signalised junction is proposed at the south to serve as the junction between local distributors Roads L11, L20 & L22 and Hung Tin Road.

Improvement Option 2

3.6.2.3 The layout of TSWW Interchange Improvement Option 2 is presented on **Appendix 3.2**. In this option, in addition to providing a local road connection between the Tin Shui Wai / Hung Shui Kiu areas and the YLS PDA, the design objective is to provide direct slip road connections between the YLS PDA and the YLH. The planning concept is to encourage part of the heavy vehicles in Zone 1 to access YLH via TSWW Interchange and hence minimising through traffic to the residential areas at the east.

3.6.2.4 Despite this, it is acknowledged that converting a 3-leg trumpet interchange to a 4-leg full-cloverleaf interchange serving 3 new towns would involve much significant land and cost. The location is impractical to house a full-colverleaf interchange. The design of improvement option 2 therefore only focuses on the connectivity with YLH westbound.

3.6.2.5 For traffic on YLH eastbound, it can access YLS PDA either via a U-turn at the diamond interchange on Hung Tin Road, or via the TYST Interchange further

downstream. The design of the TYST Interchange Improvement Works is also taking this factor into account.

3.6.2.6 For traffic on YLH westbound, additional bifurcating ramps are added to allow the traffic to access an at-grade roundabout. The at-grade roundabout provides connection between YLS Roads L11, L20 and L22, Hung Tin Road and YLH westbound. There are five arms to this junction and hence a roundabout instead of a signalised junction is more appropriate.

3.6.2.7 In order not to avoid the free flow of the traffic to the Tin Shui Wai direction, the off-slip road from YLH westbound to Tin Shui Wai is realigned to a depressed road / underpass, so as to provide grade separation with the at-grade roundabout. This bypass slip road involves a new underpass structure beneath YLH to be constructed via lane-by-lane temporary traffic management scheme. The continuously covered section of the slip road is limited to below 230m.

3.6.2.8 Segregated left-turning lane is also proposed on the roundabout to provide free flow of the on-slip road from Tin Shui Wai to YLH westbound.

3.6.2.9 A footbridge is proposed above the at-grade roundabout to provide pedestrian and cyclists connection between different sides of the roundabout and link up with the Tin Shui Wai / Hung Shui Kiu areas. A subway is not recommended as the level will be in conflict with the proposed underpass.

3.6.2.10 The level of the at-grade roundabout has to be tie-in with slip roads with YLH and the roundabout footprint has to be relatively flat. As such the junction level of Improvement Option 2 is lower than that in Option 1 and much lower than the existing ground level. The levels of the Roads L20 and L22 and nearby site formation also have to be lower. It is highlighted that the improvement option 2 will require relatively more site formation works, land requirement and cost, as compared to Option 1.

3.6.2.11 **Recommended Option**

3.6.2.12 Despite the higher cost and land requirement of Improvement Option 2, there is a strong planning and environmental objective to minimise the through traffic the residential areas in YLS PDA. Relatively less site formation work and less land requirement are involved in this improvement option, as compared to Option 2. As such, Option 2 is more suitable in term of new town planning concept and traffic noise mitigation.

3.6.2.13 In addition Option 2 offers a secondary access to the YLS PDA. A backup access point is beneficial in term of traffic planning. The improved TSWW Interchange can share the traffic burden to the TYST Interchange. Option 2 is more suitable in traffic noise mitigation by diverting through heavy good vehicle traffic away from the residential areas in YLS PDA.

3.6.2.14 Improvement Option 2 is therefore recommended for further development.

3.6.3 Alternatives for Tong Yan San Tsuen Interchange

3.6.3.1 **Improvement Option 1**

3.6.3.2 The layout of TYST Interchange Improvement Option 1 is presented on **Appendix 3.3**. In this option, the slip roads are reconfigured to provide full-movement connectivity between Long Tin Road, proposed YLS PDA and YLH. The existing

slip roads with Long Tin Road are maintained or realigned. New slip roads are added to connect with the proposed Road D1.

3.6.3.3 The improvement works involve a new underpass structure beneath YLH to be constructed via lane-by-lane traffic management scheme and several new elevated viaduct structures. Several existing bridge and underpass structures are to be abandoned and demolished.

3.6.3.4 The existing slip road are in tight radii of curvature. In the improvement works, the radii will be improved to at least the absolute values under current standard. However since there is land constraint surrounding the interchange, adoption of below-desirable radii is required. It is highlighted that the adopted radii of curvature are still better than the existing conditions.

3.6.3.5 The off-slip ramp from YLH eastbound is relocated westward. Headroom above TYST Road is maintained.

3.6.3.6 Lam Yu Road has to be diverted to provide space for 2-lane on-slip merging to YLH.

3.6.3.7 **Improvement Option 2**

3.6.3.8 The layout of TYST Interchange Improvement Option 2 is presented on **Appendix 3.3**. This option shares similar layout and design with Option 1. The exception is that a slip road from TYST Interchange to Road L11 is added. The bypass slip road goes beneath the on-slip ramp to YLH westbound in the form of depressed road and underpass structure.

3.6.3.9 The additional slip road provides a direct connection from YLH eastbound and westbound to Zone 1 Road L11. This helps in diverting heavy vehicles of logistics / industrial use away from the residential areas in Zone 2.

3.6.3.10 A direct connection from Zone 1 to YLH eastbound is not provided due to conflict with other more important slip roads. The traffic from Zone 1 can either enter YLH eastbound via the diamond interchange in Hung Tin Road to TSWW Interchange or via Road D1 to TYST Interchange.

3.6.3.11 **Recommended Option**

3.6.3.12 The addition of the bypass slip road to Road L11 involves addition cost, but better fulfil the planning intension to divert heavy vehicles away from residential areas. The additional cost is relatively small compared to the overall capital cost. Option 2 is better in term of planning and it is more suitable in traffic noise mitigation by diverting through heavy good vehicle traffic away from the residential areas in YLS PDA.

3.6.3.13 Improvement Option 2 is recommended for further development.

3.7 Construction Programme

3.7.1.1 It is anticipated that the new town development will be commissioned in phases. The construction works is targeted to commence in Year 2020 and are summarised in **Table 3.5** below.

Table 3.5 Works Packaging for the YLS Development

Works Package	Land Clearance & Construction Period	Description of Works
Stage 1 Works - Phase 1a) Initial Housing Site / Road Improvement Works	2020 to 2027	• Decontamination and Site Formation Works • Internal Road, Pipe works and Landscaping • Nullah Decking and Road Works for Kung Um Road/Kiu Hing Road and New Local Distributor to Tai Kei Leng Rd
Stage 1 Works - Phase 1b) Initial Housing Site / Resite Village House	2020 to 2029	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Re-site Village House Construction (Zone 1 and Zone 2) • Construction of Public Housing
Stage 2 Works - Phase 2a) Multi-Storey Building/ Open Storage Site (To the South West of TSWW Interchange)	2022 to 2028	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Construction of Multi-storey Buildings • Preparation of Open Storage Site
Stage 2 Works - Phase 2b) Northern Sites along Kung Um Road, Kung Um Road Improvement and Reclaimed Water Service Reservoir	2023 to 2033	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Tin Tsuen Channel Revitalisation • Kung Um Road/Kiu Hing Road/Wong Nai Tun Tsuen Road Improvement/ Yuen Long Nullah Revitalisation and Landscaping • Reclaimed Water Service Reservoir and Service Road • Re-site Village House Construction • Tin Tsuen Channel Revitalisation • Construction of Private Housing • Construction of Public Housing • Construction of Government Facilities
Stage 2 Works - Phase 2c) External Roads (Tong Yan San Tsuen Interchange Improvement, Shap Pat Heung Interchange Improvement)	2025 to 2032	• Decontamination and Site Formation Works • TYST Interchange Improvement and Landscaping • Yuen Long West Nullah Revitalisation • Re-site Village House Construction (in Zone 3) • Shap Pat Heung Interchange Improvement and Landscaping
Stage 2 Works - Phase 2d) Sewage Treatment Works	2023 to 2029	• Decontamination and Site Formation Works • Sewage Treatment Works • Pumping Station
Stage 2 Works - Phase 2e) Multi-Storey Buildings (Remaining)	2025 to 2033	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Construction of Multi-storey Buildings
Stage 3 Works - Phase 3) Southern Sites Along Kung Um Road, Kung Um Road Improvement	2031 to 2038	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Hillside River Corridor & Retention Lake • Construction of Private Housing • Construction of Public Housing • Construction of Government Facilities
Stage 4 Works - Phase 4a) Sites in Tong Yan San Tsuen and Internal Roads	2032 to 2038	• Decontamination and Site Formation Works • Internal Roads, Pipeworks, Utilities and Landscaping • Retention Tank/Lake • Construction of Private Housing • Construction of Government Facilities
Stage 4 Works - Phase 4b)	2032 to 2038	• Decontamination and Site Formation Works

Works Package	Land Clearance & Construction Period	Description of Works
TSWW Interchange Improvement		• TSWW Interchange Improvement and Landscaping